



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

SAP ABAP RICEF OBJECTS

The domain of the Project:
SAP ABAP

Team Mentor (and their designation):
Mr. Santhosh Pal

Team Member
Mr. Nanda kumar S

Period of the project
January 5th 2026 to July 5th 2026



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Declaration

The project titled "SAP ABAP RICEF OBJECTS" has been mentored by Santhosh Pal, organised by SURE Trust, from May 2026 to July 2026, for the benefit of the educated unemployed rural youth for gaining hands-on experience in working on industry relevant projects that would take them closer to the prospective employer. I declare that to the best of my knowledge the members of the team mentioned below, have worked on it successfully and enhanced their practical knowledge in the domain.

Team Member:

Mr. Nanda kumar S

Signature

Mentor's Name: Santhosh Pal

Company Name: Infosys Limited

Prof. Radhakumari

Executive Director & Founder

SURE Trust



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Table of contents

1. Executive summary
2. Introduction
3. Project Objectives
4. Methodology & Results
5. Social / Industry relevance of the project
6. Learning & Reflection
7. Future Scope & Conclusion



Executive Summary

This report presents the work carried out during a six-month SAP ABAP internship undertaken with SURE Trust, under the mentorship of Mr. Santhosh Pal, between May 2026 and July 2026. The project, titled "SAP ABAP RICEF Objects," focused on gaining hands-on, industry-relevant experience across the five core categories of SAP ABAP development: Reports, Interfaces, Conversions, Enhancements, and Forms (RICEF).

Over the course of the internship, Classical, Interactive, and ALV Reports were built to retrieve and display SAP data; an Excel-to-SAP interface was implemented for bulk data exchange; BDC and SHDB-based conversions were used to automate transactions such as MM01 and FI01; Customer Exits and BADIs were explored for enhancing standard SAP functionality; and SAP Script/Smart Forms were designed to generate professional invoices and receipts. Together, these deliverables provided practical exposure to end-to-end ABAP development and reinforced the technical foundation needed for a career in SAP consulting and development.



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Introduction

SURE Trust conducts internship programs for educated, unemployed rural youth, giving them structured, industry-relevant training and mentorship in fields such as SAP, so they are better placed to compete for jobs. As a graduate of Electronics and Communication Engineering, this internship provided a first structured entry into enterprise software development, specifically SAP ABAP, which is one of the most widely used programming languages for customising and extending SAP ERP systems.

The internship was organised around the RICEF framework, a standard classification used across the SAP ecosystem to describe the five major types of custom development objects: Reports, Interfaces, Conversions, Enhancements, and Forms. Learning was carried out through guided sessions with the mentor, self-paced practice on SAP tables and transaction codes, and the development of small working programs for each RICEF category. This report documents the objectives, methodology, outcomes, and key learnings from that experience.



Project Objectives

- To understand Reports in SAP ABAP for retrieving and displaying business data from SAP tables.
- To learn Interfaces for transferring external Excel data into SAP systems.
- To implement Conversions using BDC and SHDB for automating SAP transactions.
- To understand Enhancements such as Customer Exits, SMOD, CMOD, and BADI for SAP customisation.
- To create Forms using Smart Forms for generating invoices, receipts, and formatted business documents.



Methodology and Results

Methods / Technology Used

The project was built entirely around the RICEF classification of SAP ABAP development objects. Classical, Interactive, and ALV Reports (using the REUSE_ALV_GRID_DISPLAY function module) were used to retrieve and present data from the MKPF and MSEG material document tables. Interfaces were built to exchange data between Excel and SAP using GUI_DOWNLOAD and GUI_UPLOAD style techniques. Conversions were implemented using the BDC Session Method and Call Transaction Method, driven by SHDB recordings of the MM01 (Material Creation) and FI01 (Bank Master Creation) transactions, and later using the BAPI BAPI_PO_CREATE1 for direct, validation-based purchase order creation. Enhancements were explored through Customer Exits and BADIs (Business Add-Ins) to add custom logic to standard SAP programs without modifying the original code. Forms were built using SAP Smart Forms, with Header, Main Window, and Footer sections designed to generate invoice-style business documents.

Tools / Software Used

- SAP GUI (SAP Logon) – for accessing the SAP system and executing transactions
- ABAP Editor (SE38) – for writing and testing ABAP report programs
- ABAP Dictionary (SE11) and Data Browser (SE16/SE16N) – for viewing table structures and data
- SHDB (Transaction Recorder) – for recording BDC sessions
- SAP Smart Forms (SMARTFORMS) and SSF_FUNCTION_MODULE_NAME – for form design and generation
- Microsoft Excel – as the external data source for interface and conversion programs

Data Collection Approach

Data used during development was sourced directly from standard SAP tables such as MKPF (Material Document Header) and MSEG (Material Document Segment), retrieved using SELECT statements within the ABAP reports. For the Interface and Conversion objects, sample business data (material master records, vendor and bank details) was prepared in Excel spreadsheets and uploaded into the SAP system, simulating a real-world bulk data migration scenario.

Project Architecture

Each RICEF object followed a consistent architecture: a selection screen (or file input) gathers the user's request parameters, an ABAP program retrieves or receives the relevant data, business logic processes and validates that data, and the result is either displayed (Reports), written back into SAP (Interfaces/Conversions), triggered as a customisation hook (Enhancements), or rendered as a formatted document (Forms). For example, in the ALV report, a selection screen filters material documents by number; the report reads MKPF and MSEG, builds a field catalog, and calls REUSE_ALV_GRID_DISPLAY, with a drill-down (hotspot) from the header list into the corresponding line-item detail list. In the Excel-to-SAP conversion, data is first converted from a flat Excel upload into



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an internal table using TEXT_CONVERT_XLS_TO_SAP, mapped into BAPI structures, and posted using BAPI_PO_CREATE1, followed by a BAPI_TRANSACTION_COMMIT to save the result.

Final Project Working Screenshots

ALV Report: Material Document Header List

SAP				
MATERIAL_D...	MATERIAL_D...	TRANSACTION...	DOC_TYPE	USER_NAME
0000000001	2000		AB	ABC
0000000001	2023	WE	WE	SAR_014
0000000001	2024	WE	WE	ABAP12
0000000002	2024	WE	WE	ABAP12
0000000003	2024	WE	WE	ABAP12
0000000004			PR	ABAP12
0000000004	2024	WE	WE	ABAP12
0000000005	2024	WE	WE	ABAP12
0000000006	2024	WE	WE	ABAP12
0000000021	2020	WA		NKS15
0000000021	2024	WE	WE	73USERS5
0000000021	2026	DB		NN
0000000022	2024	WL	WL	73USERS5
0000000023	2024	WE	WE	73USERS5
0000000024	2024	WL	WL	73USERS5
0000000025	2024	WE	WE	73USERS5
0000000026	2024	WE	WE	73USERS5
0000000027	2024	WE	WE	73USERS5
0000000027	2025	IN		USER15
0000000028	2024	WA	WA	73USERS5
0000000029	2024	WE	WE	73USERS5
0000000030	2024	WA	WA	73USERS5
0000000031	2024	WE	WE	73USERS5
0000000032	2024	WA	WA	73USERS5
0000000033	2024	WA	WA	73USERS5
0000000034	2024	WA	WA	73USERS5
0000000035	2024	WA	WA	73USERS5
0000000041	2024	WL	WL	73USERS5
0000000042	2024	WE	WE	73USERS5
0000000043	2024	WL	WL	73USERS5
0000000044	2024	WE	WE	73USERS5
0000000045			BS	BTH
0000000045	2024	WL	WL	73USERS5
0000000046	2024	WL	WL	73USERS5
0000000047	2024	WE	WE	73USERS5
0000000048	2024	WL	WL	73USERS5
0000000049	2024	WE	WE	73USERS5

Fig 1 – ALV grid output showing material document header data (MKPF)

The report program ZNKS_ALV_MAJORPROJ_1 selects material document data from MKPF within a user-specified range and displays it using an ALV Grid. Clicking a row (hotspot on the Material Document field) triggers the USER_COMMAND routine, which reads the corresponding line items from MSEG and displays them in a secondary ALV list, giving an interactive drill-down experience. The report also includes a "Download Excel" pushbutton that exports the selected data to a local file using GUI_DOWNLOAD.



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Report Source Code (excerpt)

```
rt      ZNKS_INTERACTIVE_ALV_MAJORPROJ Inactive
2      *& Report  ZNKS_INTERACTIVE_ALV_MAJORPROJ
3      *&
4      *&-----
5      *&
6      *&-----
7      *&-----
8
9      REPORT ZNKS_INTERACTIVE_ALV_MAJORPROJ.
10     *DECLARE TABLE
11     TABLES: MKPF,MSEG.
12
13     *CREATE TYPES
14     TYPES: BEGIN OF ty_mkpfi,
15             MBLNR TYPE MKPF-mblnr,
16             MJahr TYPE MKPF-mjahr,
17             VGART TYPE MKPF-vgart,
18             BLART TYPE MKPF-blart,
19             USNAM TYPE MKPF-usnam,
20     END OF ty_mkpfi.
21
22     DATA: IT_MKPF TYPE TABLE OF ty_mkpfi,
23            WA_MKPF TYPE ty_mkpfi.
24
25     *SELECT SCREEN AND FETCH DATA
26     SELECT-OPTIONS: SO_MBLNR FOR MKPF-mblnr.
27
28     START-OF-SELECTION.
29     SELECT MBLNR
30            MJahr
31            VGART
32            BLART
33            USNAM
34     INTO TABLE IT_MKPF FROM MKPF WHERE MBLNR IN SO_MBLNR.
35
36     *DECLARE ALV FIELD CATLOG
37     TYPE-POOLS: SLIS.
38     DATA: IT_FCAT TYPE SLIS_T_FIELDCAT_ALV,
39            WA_FCAT TYPE SLIS_FIELDCAT_ALV.
```

Fig 2 – ABAP editor view of the interactive ALV report program

Excel Interface – Material Master Upload (MM01)

MM01

Create Material (Initial Screen)

Select View(s)	Org. Levels	Data
Material Num		
Industry sector		
Material Type		
Change Number		
Copy from...		
Material		

Fig 3 – MM01 Create Material initial screen used as the target transaction



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A	B	C	D	E	F
MATERAIL	IND SEC	MAT TYPE	UCOM	DISCRIPTION	MAT GROUP
MAT9535	P	ZPAC	EA	RAW MAT 163	001
MAT9536	P	ZPAC	EA	RAW MAT 164	001
MAT9537	P	ZPAC	EA	RAW MAT 165	001

Fig 4 – Sample Excel data sheet used as the source file for the interface

Material master data prepared in Excel (material number, industry sector, material type, unit of measure, description, and material group) was converted into an internal table using TEXT_CONVERT_XLS_TO_SAP and posted into SAP through the BAPI-based processing shown in the accompanying purchase-order/material program, demonstrating a working Excel-to-SAP data conversion flow.

Smart Forms – Invoice Layout and Output

```

ABAP Editor: Change Report ZNKS_MAJPROJ_SMARTFORMS

Report ZNKS_MAJPROJ_SMARTFORMS Inactive

19 DATA: it_mseg TYPE STANDARD TABLE OF znks_majproj_smart,
20 wa_mseg TYPE znks_majproj_smart,
21 lv_fm_name TYPE rs36l_fnam.
22 DATA: lv_dmbtr TYPE string.
23 DATA: gv_total TYPE p DECIMALS 2,
24 gv_total_char TYPE char20,
25 gv_words TYPE char255.
26
27 DATA: lv_spell TYPE spell.
28
29 PARAMETERS: p_mblnr TYPE mseg-mblnr.
30 SELECTION-SCREEN PUSHBUTTON /1(20) btn1 USER-COMMAND EXL.
31
32 INITIALIZATION.
33   btn1 = 'Download Excel'.
34   AT SELECTION-SCREEN.
35
36 CASE sy-ucomm.
37 WHEN 'EXL'.
38   PERFORM download_excel.
39
40 ENDCASE.
41
42 START-OF-SELECTION.
43
44 DATA: BEGIN OF ls_mseg_db,
45   mblnr TYPE mseg-mblnr,
46   mjahr TYPE mseg-mjahr,
47   bwart TYPE mseg-bwart,
48   werks TYPE mseg-werks,
49   dmbtr TYPE mseg-dmbtr,
50   END OF ls_mseg_db.
51
52 DATA: lt_mseg_db LIKE TABLE OF ls_mseg_db.
53
54 SELECT mblnr
55        mjahr
56
```

Fig 5 – ABAP program ZNKS_MAJPROJ_SMARTFORMS calling the Smart Form via SSF_FUNCTION_MODULE_NAME



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SAP Form Builder: Change Form ZNKS_MAJPROJ

Form Painter

Form: ZNKS_MAJPROJ, Active

Meaning: New Form

General Attributes | Output Options

Created by: ABAP75, Changed by: 73USERS
 Date: 05.06.2026, Date: 06.07.2026
 Time: 13:59:31, Time: 11:56:19

Package: \$TMP

Language Attributes

Language: EN

Translate

☒ Into All Languages
☐ Into Selected Languages
☐ Do Not Translate

Access

☐ Restricted Language Control
☐ Only Process Installed Languages

Form Structure:

- Global Settings
 - Form Attributes
 - Form Interface
 - Global Definitions
- Pages and Windows
 - %PAGE1 New Page
 - %TEXT14 New Text 14
 - COMPANY COMPANY
 - %TEXT13 New Text 13
 - GRAPHICS GRAPHICS
 - MAIN Main Window
 - %TABLE1 New Table 1

Header

- %ROW1 New Line 1
 - MAT_DOC MAT_DOC
 - MAT_DOC_YEAR mat
 - MOVEMENT_TYPE m
 - PLANT plant
 - AMOUNT AMOUNT

Main Area

- %ROW2 New Line 2
 - MAT_DOC1 mat_doc1
 - MAT_DOC_YEAR1 M
 - MOVEMENT_TYPE1 n
 - PLANT1 plant1
 - AMOUNT1 amount1

Footer

- %ROW3 New Line 3
 - AMTWORD amtword
 - TOTAL_1 total_1

Fig 6 – Smart Form Builder layout for ZNKS_MAJPROJ, showing Header, Main, and Footer windows

Print Preview of LP01 Page 00001 of 00001

Archive | Print and Archive

UNIVERSAL 15TH MAIN 10
 ROAD STREET ,MANGLORE

material_doc	material_doc_year	movement_type	plant	AMOUNT
65	2024	101	MMI1	200,00
Amount In Words: TWO HUNDRED				TOTAL AMOUNT
				200,00

SIGNATURE

Fig 7 – Final print preview output of the Smart Form invoice, including amount-in-words

The Smart Form ZNKS_MAJPROJ was designed with a Header window (company name, logo, and address), a Main window (material document, year, movement type, plant, and amount), and a Footer window (total amount, amount in words using the SPELL_AMOUNT function module, and a signature line), producing a clean, print-ready business document from live SAP data.

Project GitHub Link

[Add your project GitHub repository link here]



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Social / Industry Relevance of the Project

SAP remains one of the most widely used enterprise resource planning (ERP) platforms across manufacturing, retail, banking, and public-sector organisations worldwide, and ABAP is its core development language. Skills in RICEF objects – Reports, Interfaces, Conversions, Enhancements, and Forms – are directly applicable to real SAP implementation and support projects, making this project highly relevant to industry hiring needs in the SAP ecosystem.

For educated rural youth, as targeted by SURE Trust, structured exposure to industry-grade tools such as SAP is especially valuable: it bridges the gap between academic knowledge and the practical, tool-specific skills that employers in IT services and enterprise consulting look for. The BDC/BAPI-based data conversion techniques learned here are commonly used during SAP implementation and data-migration projects, while the ALV reporting and Smart Forms skills map directly to reporting and documentation needs that almost every SAP-run organisation has. This project therefore contributes both to individual employability and to a broader goal of building local, industry-ready SAP talent.



Learning and Reflection

New Learnings

- Gained hands-on knowledge of the RICEF framework, including Classical, Interactive, and ALV reporting techniques using REUSE_ALV_GRID_DISPLAY.
- Learned to build Excel-to-SAP interfaces and to automate data uploads using BDC (SHDB recording, Call Transaction/Session methods) and BAPIs such as BAPI_PO_CREATE1.
- Understood SAP enhancement techniques, including Customer Exits and BADIs, and when each is appropriate for customising standard SAP behaviour.
- Built a working Smart Form invoice layout, including Header/Main/Footer window design and amount-in-words conversion using SPELL_AMOUNT.
- Improved debugging skills and general SAP navigation ability (SE38, SE11, SE80, SHDB, SMARTFORMS) through repeated hands-on practice.

Overall Experience

Coming from an Electronics and Communication Engineering background, this internship was my first structured, sustained exposure to enterprise software development. Working through each RICEF object one at a time, under the guidance of Mr. Santhosh Pal, helped me move from theoretical programming concepts to practical, business-oriented problem solving. The experience strengthened my confidence in ABAP coding, gave me a much clearer picture of how SAP is actually used in industry, and reinforced my interest in pursuing a career in SAP ABAP development.



Conclusion and Future Scope

Conclusion

This internship achieved its objective of providing hands-on exposure to all five categories of the RICEF framework. Reports built understanding of data retrieval and interactive/ALV display techniques; Interfaces and Conversions provided practical knowledge of Excel integration and transaction automation using BDC and BAPIs; Enhancements introduced SAP customisation through Customer Exits and BADIs; and Forms gave direct experience of invoice and receipt generation using Smart Forms. Overall, working across these RICEF objects improved technical ability, debugging skills, SAP navigation, and understanding of real-world enterprise application development.

Future Scope

- Extend the ALV report into a full interactive dashboard with additional drill-downs, sorting, and export options.
- Explore newer SAP technologies such as OData services (SEGW), CDS Views, and the RAP model to modernise the RICEF objects built here for SAP S/4HANA and Fiori.
- Build on the Module Pool (Tabstrip) programming practised alongside this project to create a complete custom transaction screen combining these RICEF objects.



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